

React Interview Prep

Topic 10 — useContext: Solving Prop Drilling

BEGINNER → INTERMEDIATE → EXPERT

■ Covers

Prop drilling · createContext · Provider · useContext · Re-render problem · Custom hook pattern

■ Series

React Interview Prep — Session 10

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1. The Problem — Prop Drilling

Passing props through components that don't need them, just to reach a deeply nested child.

```
function App() {
  const user = { name: 'Akshay', role: 'Developer' };
  return <Header user={user} />; // passed down
}

function Header({ user }) {
  // ■ Header doesn't need user — just a middleman
  return <UserAvatar user={user} />;
}

function UserAvatar({ user }) {
  // Finally used 2 levels deep
  return <img src={`/avatars/${user.name}.png`} />;
}
```

■ **Prop drilling becomes painful at 3+ levels with multiple values. useContext eliminates the middlemen — any component reads directly.**

2. The 3 Steps of useContext

Step 1 createContext()



Step 2 Provider



Step 3 useContext()

Step 1: createContext() — Create the notice board

```
import { createContext } from 'react';

const UserContext = createContext(null); // null = default value

export default UserContext;

// Convention: name it SomethingContext. Export for other files.
```

Step 2: Provider — Put data on the board

```
import UserContext from './UserContext';

function App() {

  const user = { name: 'Akshay', role: 'Developer' };

  return (

    // no prop needed anymore ■

    // can read user directly ■

  );

}
```

Step 3: useContext() — Read from the board anywhere

```
import { useContext } from 'react';

import UserContext from './UserContext';

function UserAvatar() {

  const user = useContext(UserContext); // reads directly ■

  return <img src={avatarUrl} />;

}

// No props needed. No middlemen. Direct access.
```

3. Full Example — Theme Toggle

A real working example: theme read AND updated from a deeply nested component.

```
import { createContext, useContext, useState } from 'react';
const ThemeContext = createContext('light');
export default function App() {
  const [theme, setTheme] = useState('light');
  return (
    <ThemeContext.Provider value={{ theme, setTheme }}>
    <Page />
    </ThemeContext.Provider>
  );
}
function Page() {
  return <div><Navbar /><Content /></div>;
  // Page doesn't receive OR pass any props ■
}
function Navbar() {
  const { theme, setTheme } = useContext(ThemeContext);
  return (
    <nav style={{ background: theme === 'dark' ? '#333' : '#fff' }}>
    <button onClick={() =>
      setTheme(t => t === 'light' ? 'dark' : 'light')}>
      Toggle Theme
    </button>
    </nav>
  );
}
function Content() {
  const { theme } = useContext(ThemeContext);
  return (
    <main style={{ color: theme === 'dark' ? '#fff' : '#000' }}>
    Hello Akshay!
    </main>
  );
}
```

4. The Re-render Problem — Expert

Every consumer re-renders when the Provider value changes. If value is a new object every render — all consumers re-render unnecessarily.

■ Wrong

```
// ■ New object on every render
// count change → new value object
// → ALL context consumers
re-render

function App() {

  const [theme, setTheme] =
    useState('light');

  const [count, setCount] =
    useState(0);

  return (

    <ThemeContext.Provider
      value={{ theme, setTheme }}>
      // new object every render ■
    </ThemeContext.Provider>

  );
}
```

■ Correct

```
// ■ useMemo stabilizes the value
// count change → same value
object
// → consumers do NOT re-render

function App() {

  const [theme, setTheme] =
    useState('light');

  const [count, setCount] =
    useState(0);

  const val = useMemo(
    () => ({ theme, setTheme }),
    [theme] // only new obj when theme
    changes
  );

  return (

    <ThemeContext.Provider>

  );
}
```

■ **Always wrap Provider value in useMemo when it's an object. Otherwise every unrelated state change triggers all consumers.**

5. Custom Hook Pattern — Production Best Practice

Instead of importing the context + useContext everywhere, wrap it in a custom hook. This is the pattern used in real production apps.

```
const AuthContext = createContext(null);
// Custom hook – hides implementation, adds safety check
export function useAuth() {
  const context = useContext(AuthContext);
  if (!context) {
    throw new Error('useAuth must be inside AuthProvider');
  }
  return context;
}

// Provider component – clean wrapper
export function AuthProvider({ children }) {
  const [user, setUser] = useState(null);
  const login = useCallback((u) => setUser(u), []);
  const logout = useCallback(() => setUser(null), []);
  const value = useMemo(
    () => ({ user, login, logout }), [user, login, logout]
  );
  return (
    <AuthContext.Provider value={value}>
      {children}
    </AuthContext.Provider>
  );
}

// In any component – super clean, no context import needed
function Navbar() {
  const { user, logout } = useAuth(); // ■
  return <button onClick={logout}>{user.name}</button>;
}
```

■ Custom hook pattern = cleaner API + safety check + hidden implementation. This is the pattern used in real codebases — always prefer this over raw useContext.

6. Context vs Props vs Redux/Zustand

Situation	Best choice
Data needed 1-2 levels deep only	Props — simplest, most explicit
Global UI: theme, language, auth, modals	useContext — built-in, no extra library
High-frequency updates (mouse pos, scroll)	Local state — context re-renders all consumers
Complex state with many update sources	Redux or Zustand
Large teams needing strict patterns	Redux — enforces discipline
Small-medium app needing global state fast	Zustand — minimal, fast to set up

■ *useContext is NOT a replacement for Redux/Zustand. It solves prop drilling. Redux/Zustand solve complex state management.*

7. Interview Q&A;

Q

Beginner

Q1. What is useContext and why does it exist?

→ useContext solves prop drilling — passing data through many components that don't need it just to reach a deeply nested one. createContext creates a shared data store, Provider puts data into it, and useContext reads it from any component in the tree without prop chains.

Q

Beginner

Q2. What are the 3 steps of useContext?

→ First — createContext() to create the context object. Second — wrap components with Context.Provider and set its value prop. Third — call useContext(ContextName) inside any component that needs the data. No prop chains needed.

Q

Beginner

Q3. What is prop drilling?

→ Prop drilling is when data is passed through multiple intermediate components that don't actually use it — just to reach a deeply nested component that does. It creates tight coupling and makes refactoring painful. useContext eliminates these middlemen.

Q

Intermediate

Q4. What is the re-render problem with Context?

→ Every time the Provider value changes, ALL components consuming that context re-render — even if the specific data they use didn't change. Fix: wrap the value in useMemo so a new object is only created when relevant data actually changes, not on every parent re-render.

Q

Intermediate

Q5. What is the custom hook pattern with Context?

→ Instead of importing the context and useContext in every file, create a custom hook like useAuth() that wraps useContext internally. This hides implementation, lets you add a safety check for missing Provider, and gives consumers a clean API — just call useAuth() anywhere.

Q

Intermediate

Q6. When should you NOT use Context?

→ Avoid Context for high-frequency values like mouse position or scroll that change constantly — every consumer re-renders on every change causing performance issues. Also avoid it when state is complex with many interdependencies — use Redux or Zustand instead.

Q

Expert

Q7. What is the difference between Context and Redux/Zustand?

→ Context is built-in and solves prop drilling for relatively stable values. It has no middleware, no DevTools, no time-travel. Redux enforces strict unidirectional flow with full DevTools and middleware support — great for large teams. Zustand is minimal with direct state updates — great for medium apps. Context alone is not a state manager.

Q**Expert****Q8. Can you have multiple contexts in one app?**

→ Yes — you can nest as many Providers as needed. Common pattern: AuthContext for user, ThemeContext for dark/light, CartContext for e-commerce. Each component reads only the context it needs. Splitting contexts also prevents unnecessary re-renders — a theme change won't re-render auth consumers.